

THE WILD TOOTH THEOREM

El Diente Salvaje — the prime the Bend left outside, tamed by its own ledger; the deep towers a kingdom of depth mod 4

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Companion to THE WATERMARK THEOREM, THE DOUBLE LADDER THEOREM, THE SWEET LIE THEOREM and THE BEND THEOREM (same repository).

Computations: MacBook Air M2, 8 GB RAM, single thread, 25% CPU.

The Bend Theorem proved where and why the discriminant profile law of Fermat surfaces bends at odd primes, and closed its statement with a parenthesis: $p = 2$, even degree — both channels open and wild — the even halves separate kingdom, outside this document. This document enters that kingdom and tames the animal. The prime 2 is wild for two reasons, both now named: over $\mathbf{F}_2$ every power-sum channel is open at once — $\sum t^j \equiv 1$ for every $j \geq 1$, including the odd channels $k = 1, 3$ that no odd prime ever sees — and, deeper, the polynomial calculus itself FOLDS: $t^2 \equiv t$, so quadratic and linear are the same function and the odd kingdom's bookkeeping collapses. A folded frame with every door open is not chaos; it is a smaller, stranger room. What survives the collapse is a sharper law than the wildness suggested: one constant for every squarefree even degree, one constant for every deep tower of two — the latter NEGATIVE, a rank deficit no odd prime ever produced — one one-line ledger that explains both, and the odd teeth besides, and an anatomy in which **every dimension carries a name**, closed in the same day the document was written.

0. STATEMENT

Let S_m be the Fermat surface of even degree m , G the SSvL line Gram, σ the shadow map onto the twelve channels, and $\mathbf{b}(2) = \text{rank}_{\{\mathbf{F}_2\}}(G) - 12(m-3)$ the bend constant at 2.

Theorem (the Wild Tooth).

- **[Squarefree law, FIRM]** For every squarefree even m : $\mathbf{b}(2) = 14$, with uniform anatomy: radical of dimension 7 — five surviving cross-family **mass ties** and two cross-family **parity ties**, every dimension named (§2) — and shadow corank 15.
- **[Tower law, FIRM]** For $m = 2^e$, $e \geq 3$: $\mathbf{b}(2) = -6$ — a NEGATIVE bend, rank below the $12(m-3)$ baseline — with constant anatomy: radical 21 (parity-polynomial 15, plus six **second-bit ties**, §4), shadow corank 21. The ground floor $m = 4$ is the lone exception: $\mathbf{b}(2) = +6$ (corank already 21, radical still 9).
- **[The universal ledger — the crown]** All bend constants, at every prime and every degree measured, obey one line:

$$\mathbf{b}(p) = 36 - \text{corank}_p(\sigma) - \dim(\text{radical}_p),$$

with $36 = 12 \text{ channels} \times 3 \text{ polynomial degrees}$ the fixed baseline. Verified seven of seven: $(m,p) = (15,5): 18+16 \rightarrow \mathbf{b} = 2 \cdot (13,13): 18+18 \rightarrow 0 \cdot (6,2): 15+7 \rightarrow 14 \cdot (10,2): 15+7 \rightarrow 14 \cdot (4,2): 21+9 \rightarrow +6 \cdot (32,2): 21+21 \rightarrow -6 \cdot$ and the odd Bend's accounting $\mathbf{b} = 18 - \text{radical recovered entire as the corank-18 special case. Odd primes never move the corank; the wild tooth moves BOTH terms.}$

1. THE TWO FACES OF WILDNESS

(i) All channels open. Over $\mathbf{F}_2$, $\sum t^j \equiv 1$ for every $j \geq 1$ — the $(p-1) \mid k$ criterion of the Bend admits everything, including the odd channels invisible to every odd prime. The Bend's kernel structure (the generalized circulant $K(t,t') = \kappa(\alpha t + \beta t' \bmod m)$, proven at general m) applies verbatim; what changes is that no door is ever shut.

(ii) The fold. As functions on $\mathbf{F}_2$, $t^2 \equiv t$: per channel the polynomial space collapses to {constant, parity}. The odd kingdom's generic frame — 18 radical identities, 18 shadow relations, both polynomial of degree ≤ 2 — does not reduce mod 2; it FOLDS, and the wild tooth's arithmetic is the arithmetic of that folding, at two levels.

2. THE FOLD AT THE RADICAL LEVEL — every dimension named

Mod 2 the certified linear (6) and polarization (3) frames land in one parity sector of rank **5** — four collisions, polarization $\equiv$ linear by the fold. The true mass frame (extracted from two odd kingdoms, $m = 15$ p5 and $m = 13$ p13: dimension 9 at both, identical integral span, entries $\{0, \pm 1\}$ — each generator a cross-family mass tie) keeps full rank 9: the constant sector does not fold. Death census with full tail freedom, byte-identical at $m = 6$ and $m = 10$: **mass deaths 4, survivors 5 · parity deaths 4, survivor 1** — and pure parity content in the radical is ZERO: nothing of the parity sector survives without a constant tail welded on.

The seventh dimension — and with it the full mixed pair — is the even kingdom's own inhabitant, now named: the **cross-family parity ties**, the parity analogue of the odd mass ties. The all-directions parity mass of two families agrees up to a constant tail, and the tails are exact and m -independent:

$$\text{allpar}(f_0) + \text{allpar}(f_1) \equiv 1[f_0:k-1] + 1[f_0:k+1] + 1[f_1:k], \text{allpar}(f_0) + \text{allpar}(f_2) \equiv 1[f_0:k] + 1[f_2:k],$$

with the third tie their sum — cyclic consistency exact. Verified at THREE kingdoms ($m = 6, 10, 14$, byte-identical tails); the asymmetry of the tails records the unit shift of the family-0/family-2 incidence rule. **Census closed: 7 = 5 mass ties + 2 parity ties, nothing unnamed.**

3. THE FOLD AT THE CORANK LEVEL — the three collisions, named

The 18 shadow relations (gated at the odd control $m = 15$, $p = 5$: all 18 in the left kernel of σ , rank 18 = corank, full span — no mod-2 reading trusted before that gate fired green) are, per family: three constant ties (the four direction-masses agree), two linear relations ($-k+l+(k-1)$ and $-k-l+(k+l)$), and the **parallelogram identity** $(k-1)^2 + (k+l)^2 = 2k^2 + 2l^2$. Mod 2:

$$\text{PARALLELOGRAM} \equiv \text{lin}_1 + \text{lin}_2 \text{ (one collision per family, three in all: 18} \rightarrow \text{15).}$$

Verified two ways: machine extraction byte-identical at $m = 6$ and $m = 10$, and symbolically — the parallelogram's weight vector $(-2t^2, -2t^2, t^2, t^2)$ folds to $(0, 0, t, t)$, which is exactly $\text{lin}_1 + \text{lin}_2 \equiv (-2t, 0, t, t) \bmod 2$. The corank term of the squarefree ledger (15) is the folded trinity, and the bend's $14 = 36 - 15 - 7$ closes as pure bookkeeping.

4. THE TOWERS — THE MOD-4 KINGDOM, AND THE NEGATIVE BEND

At $m = 2^e$ the shadow corank is **21 = 18 - 3 + 6**: the same three fold-collisions, PLUS six new relations (two per family) that exist only when $4 \mid m$ — **mod-4 carry windows**, identities of residue indicators, e.g. per family the complementary pair $\text{ind}[k \in \{0, 1\}] + \text{ind}[l \in \{1, 2\}] + \text{ind}[k-l=1] + \text{ind}[k+l=0] \equiv 0$ and its bit-flipped twin — verified by hand on all 16 points of $(\mathbf{Z}/4)^2$, carry arithmetic of the half-residues, structure no odd kingdom can possess.

The radical meanwhile grows from 9 ($m = 4$) to a stable 21 ($m = 8$ through 64), carrying six dimensions beyond the polynomial frame — the campaign's first radical content outside constants and parities anywhere. Those six are now identified: as cosets modulo the polynomial frame they are spanned by the **second bit of t** — the patterns $\text{ind}[t \bmod 4 \in \{2, 3\}] (= b_1)$ and

$\text{ind}[t \bmod 4 = 3]$ (= b_1 -parity) — in cross-family tie combinations, the **second-bit ties** (the cleanest: the second bit of the k -l channels of families 1 and 2, tied). Extracted at $m = 8$ and $m = 16$ with byte-equal structure, the supports doubling exactly as residue classes mod 4: at $m = 16$ the patterns still factor through $t \bmod 4$, never mod 8.

The unified statement: the deep 2-tower lives entirely at depth mod 4. The corank's six carry windows and the radical's six second-bit ties are the two faces of one phenomenon — the first Jordan layer above parity — and the measured constancy of the whole anatomy from $m = 8$ through $m = 64$ says no deeper layer ever activates. The ledger then explains the walk with no further input: $m = 4$: $36 - 21 - 9 = +6$ (the windows already exist, the second-bit layer does not yet — at $m = 4$ the quarter is a single residue and the bit degenerates); $m \geq 8$: $36 - 21 - 21 = -6$. The negative bend is corank inflation and radical inflation together overshooting the baseline — **explained, not just measured**.

A correction carried openly: an earlier session note recorded the 2-adic filtration route as half-dead after a mod-4 layer test captured nothing. The proper coset extraction shows the opposite — the beyond content IS mod-4 structure; the earlier null was an accounting artifact of the layer test. The correction, with its data, is in the campaign acta — the only grave this campaign has ever exhumed.

5. THE FIELD RECORD

Squarefree: $b(2) = 14$ at $m = 6, 10, 14$ (blind census) joining the sealed (2,22) and (2,26) — five sightings, zero exceptions, **[FIRM]** (the Auditor re-measured $m = 6, 10$ byte-exact in a second environment). Towers: $b(2) = +6, -6, -6, -6, -6$ at $m = 4, 8, 16, 32, 64$ — the deep constant **[FIRM]** ($m = 32$ re-measured with an independent from-scratch bitset eliminator: rank $342 = 348 - 6$), anatomy byte-constant through three full splits. Leak gate: $\ker(\sigma)$ pairs to zero under G at both squarefree cells — the form factors through the shadows at 2; every theorem sentence above was written after that gate, not before. The closing extraction machinery itself was gated against three sealed radical values (7, 21, 16 — two primes, three regimes) before any new object was trusted. Ledger: seven of seven, both parities, all three regimes.

6. HONEST LEDGER

Proven / identified, nothing below: the two faces of wildness; the universal ledger at seven stations; the fold at both levels with every collision named (radical: four; corank: three, parallelogram $\equiv \text{lin}_1 + \text{lin}_2$, verified symbolically and by machine); the squarefree anatomy with every radical dimension carried by an explicit identity verified at three kingdoms; the tower anatomy with both inflations identified and the mod-4 depth statement; the mod-4 carry windows hand-verified 16/16; the true mass frame cross-kingdom certified; the leak gate paid.

Measured [FIRM]: the two field laws and their anatomies.

Scope: odd degrees obey the Bend Theorem; this document is the even half. The *discriminant* of even-degree surfaces (as opposed to the rank profile) is a different invariant and a different campaign — nothing here claims it.

7. GRAVEYARD, WITH PRIDE

The cubic even echo (registered in the acta before firing: the 2-tooth should echo $\rho(S_2) = 2$; measured global survivor $1 \neq 2$ — dead at first firing, falsifier published) · the nine-candidate within-family mass frame (the odd-kingdom control gate bit it at 7/9; its autopsy yielded the true cross-family frame, and the gate it built is now standard equipment) · the half-dead verdict on the 2-adic filtration route — **exhumed**: the only grave this campaign ever reopened, resurrected by the proper coset extraction with the correction recorded · one right-kernel-for-left-kernel script slip, self-caught on dimension reconciliation · one wrong radical convention and one naive coset

slice in the closing extraction, both self-caught against the three-seal gate before anything was trusted. *Each grave bought road — and one gave it back.*

8. PROVENANCE

Instruments: GRAM_MANOMETER_SURFACE_v4.py (ratified), the packed-bitset F_2 eliminators (towers to $n = 12288$ in seconds; $m = 32$ reproduced by a second, independent implementation), the gated shadow-extraction machinery of the closing session (radical = $\sigma(\ker G)$, calibrated against three sealed values before use), and the session extraction scripts (frames, dossiers, window identities). Seals declared before every measurement; odd-kingdom control gates before every mod-2 reading (the campaign's standing lesson, built in); this document written in crossfire by both instances of the dual protocol and merged under the Auditor's responsibility, with cross-review per house rule. All numbers from files.

References

1. M. Schütt, T. Shioda, R. van Luijk, *Lines on Fermat surfaces*. J. Number Theory **130**:9 (2010), 1939–1963. arXiv:0812.2377. (*The line intersection rules behind the Gram and the shadow map.*)
2. T. Shioda, *Some observations on Jacobi sums*. Adv. Stud. Pure Math. **12** (1987), 119–135. (*The discriminant questions; the even degrees are the half this document enters.*)
3. E. Aljovin, H. Movasati, R. Villaflor, *Integral Hodge conjecture for Fermat varieties*. J. Symbolic Computation **95** (2019), 177–184. arXiv:1711.02628. (*The published cell-by-cell elementary divisors; the campaign's gates.*)

Companion documents in this repository: THE WATERMARK THEOREM (the prime-degree discriminant) · THE DOUBLE LADDER THEOREM (the surface structure) · THE SWEET LIE THEOREM (the fourfold rank) · THE BEND THEOREM (the odd teeth; this document is its wild sibling) · CHUCHIPACHI FINDINGS MASTER (the campaign record, graveyards included). **The fifth document of the collection. PMC.**